

4.2 Creating a SQLite Database in Python

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CS XII

Chapter 4: Data and Analysis

Lecture 03 of 9

4.2 Creating a SQLite Database in Python

LEARN IT

SQLite is built into Python. The whole database is one `.db` file next to your script — no password, no port, no server. Nine steps you must be able to write from memory:

```
import sqlite3
conn = sqlite3.connect("school.db") # 1-2 driver + connection
cur = conn.cursor() # 3 cursor
cur.execute("CREATE TABLE IF NOT EXISTS students (id INTEGER, name TEXT, marks INTEGER)")
cur.execute("INSERT INTO students VALUES (1, 'Ali', 85)")
conn.commit() # 6 SAVE or the row vanishes
cur.execute("SELECT * FROM students")
rows = cur.fetchall()
for r in rows:
    print(r)
conn.close() # 9 unlock the file
```

OUTPUT

```
(1, 'Ali', 85)
```

Step	Why it matters
connect	Creates the file if it does not exist
cursor	Vehicle that carries SQL
execute	Runs one SQL command
commit	Writes INSERT/UPDATE/DELETE to disk
fetchall	Brings SELECT results into a Python list of tuples
close	Frees memory and unlocks the file

COMMON MISTAKE

Forgetting `commit()` is the classic bug: the program prints nothing next time because the insert was never saved. Forgetting `close()` can lock `school.db` so the next run fails.

REMEMBER IT

Strings inside SQL use single quotes: `'Ali'`. Python strings around the SQL use double quotes (or a triple-quoted string).

اردو میں سمجھیں

ایس کیو لائٹ پائتھن کے ساتھ آتی ہے، پورا ڈیٹا بیس ایک فائل ہے۔ ترتیب: امپورٹ، کنکٹ، کرسر، EXECUTE, COMMIT, SELECT، نہ ہو تو اندراج ضائع ہو جاتا ہے۔ COMMIT, FETCALL, CLOSE

CHECK YOURSELF

1. What does `*` mean in `SELECT *`?
2. Which call must follow `INSERT`?

Answers: 1. All columns. 2. `conn.commit()`.

Next lecture: 4.3 Pandas and DataFrames